Granisetron: Drug information - UpToDate

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Contributor Disclosures

For additional information see "Granisetron: Patient drug information" and "Granisetron: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Granisol;
- Sancuso;
- Sustol

Brand Names: Canada

- APO-Granisetron;
- JAMP Granisetron;
- NAT-Granisetron

Pharmacologic Category

- Antiemetic;
- Selective 5-HT_3 Receptor Antagonist

Dosing: Adult

Chemotherapy-induced nausea and vomiting, prevention

Chemotherapy-induced nausea and vomiting, prevention:

***General (manufacturer) dosing recommendations*:**

Oral: 2 mg once daily up to 1 hour before chemotherapy **or** 1 mg twice daily; the first 1 mg dose should be administered up to 1 hour before chemotherapy (with the second 1 mg dose 12 hours later). Administer only on the day(s) chemotherapy is administered. Continued treatment while not on chemotherapy has not been found to be useful (Ref).

IV: 10 mcg/kg within 30 minutes prior to chemotherapy; administer only on the day(s) chemotherapy is administered (Ref).

SUBQ (ER injection): Moderately emetogenic chemotherapy or anthracycline/cyclophosphamide chemotherapy: 10 mg at least 30 minutes prior to chemotherapy on day 1 (in combination with IV dexamethasone on day 1 and [for anthracycline/cyclophosphamide chemotherapy] oral dexamethasone on days 2 to 4); do not administer more frequently than once every 7 days. May also be administered in combination with a neurokinin 1 (NK_1) receptor antagonist antiemetic regimen (Ref).

Transdermal patch: Prophylaxis of chemotherapy-related emesis: Apply 1 patch at least 24 hours prior to chemotherapy; may be applied up to 48 hours before chemotherapy. Remove patch a minimum of 24 hours after chemotherapy completion. Maximum duration: Patch may be worn up to 7 days, depending on chemotherapy regimen duration (Ref).

Guideline dosing recommendations:

Highly emetogenic chemotherapy (>90% risk of emesis [eg, cisplatin, breast cancer regimens that include an anthracycline combined with cyclophosphamide]) (Ref):

Day of chemotherapy: *Administer prior* to chemotherapy **and** in combination with an NK_1 receptor antagonist, dexamethasone, and olanzapine (Ref).

IV: 1 mg or 10 **mcg**/kg as a single dose (Ref).

Oral: 2 mg as a single dose (Ref).

SUBQ: 10 mg as a single dose (Ref).

Transdermal: Apply 1 patch (Ref).

Postchemotherapy days: 5-HT_3 receptor antagonist use is **not** necessary (other components of the antiemetic regimen are administered) (Ref).

Moderately emetogenic chemotherapy (30% to 90% risk of emesis): Carboplatin-based regimens (carboplatin AUC ≥4) (Ref):

Day of chemotherapy: Administer prior to chemotherapy **and** in combination with an NK_1 receptor antagonist and dexamethasone.

IV: 1 mg or 10 **mcg**/kg as a single dose (Ref).

Oral: 2 mg as a single dose (Ref).

SUBQ: 10 mg as a single dose (Ref).

Transdermal: Apply 1 patch (Ref).

Postchemotherapy days: 5-HT_3 receptor antagonist use is not necessary (other components of the antiemetic regimen may be administered).

Moderately emetogenic chemotherapy (30% to 90% risk of emesis): Non–carboplatin-based regimens or carboplatin AUC <4) (Ref):

Note: Guidelines do not state a preference for which 5-HT_3 receptor antagonist should be used in this setting; however, palonosetron may be preferred (Ref).

Day of chemotherapy: Administer prior to chemotherapy **and** in combination with dexamethasone.

IV: 1 mg or 10 **mcg**/kg as a single dose (Ref).

Oral: 2 mg as a single dose (Ref).

SUBQ: 10 mg as a single dose (Ref).

Transdermal: Apply 1 patch (Ref).

Postchemotherapy days: 5-HT_3 receptor antagonist use is not necessary (other components of the antiemetic regimen may be administered).

Low emetogenic risk (10% to 30% risk of emesis) :

Note: Single-agent granisetron is an option for prophylaxis (Ref).

Day of chemotherapy:

IV: 1 mg or 10 **mcg**/kg as a single dose (Ref).

Oral: 2 mg as a single dose (Ref).

SUBQ (off label): 10 mg as a single dose (Ref).

Transdermal (off label): Apply 1 patch (Ref).

Postchemotherapy days: Prophylaxis is not necessary on subsequent days.

Minimal emetogenic risk (<10% risk of emesis): Routine antiemetic prophylaxis is not generally necessary (Ref).

Chemotherapy-induced nausea and vomiting, prevention: High-dose chemotherapy with stem or bone marrow transplant:

Day of chemotherapy: Administer prior to chemotherapy **and** in combination with an NK_1 receptor antagonist, dexamethasone, ± olanzapine (Ref).

IV: 1 mg or 10 **mcg**/kg as a single dose (Ref).

Oral: 2 mg as a single dose (Ref).

SUBQ (off label): 10 mg as a single dose (Ref).

Transdermal: Apply 1 patch (Ref).

Postoperative nausea and vomiting, prevention

Postoperative nausea and vomiting, prevention: IV: 0.35 to 3 mg (5 to 20 **mcg**/kg) administered before anesthetic induction or immediately before reversal of anesthesia (Ref).

Postoperative nausea and vomiting, treatment

Postoperative nausea and vomiting, treatment: IV: 0.1 to 1 mg as a single dose (Ref).

Radiation therapy–associated emesis, prophylaxis

Radiation therapy–associated emesis, prophylaxis:

High-emetogenic risk radiation therapy (total body irradiation):

Radiation day(s):

IV (off label): 1 mg or 10 **mcg**/kg once daily prior to each fraction of radiation; administer in combination with dexamethasone (Ref).

Oral: 2 mg once daily administered 1 to 2 hours prior to each fraction of radiation; administer in combination with dexamethasone (Ref).

Postradiation days:

IV (off label), Oral: The appropriate duration of therapy following radiotherapy days is not well defined; ASCO guidelines recommend continuing granisetron once daily on the day after each day of radiation (Ref).

Moderate-emetogenic risk radiation therapy (upper abdomen, craniospinal irradiation):

Radiation day(s):

IV (off label): 1 mg or 10 **mcg**/kg once daily prior to each fraction of radiation; may administer with or without dexamethasone before the first 5 fractions (Ref).

Oral: 2 mg once daily administered 1 to 2 hours prior to each fraction of radiation; may administer with or without dexamethasone before the first 5 fractions (Ref).

Low-emetogenic (brain, head and neck, thorax, pelvis) or minimal-emetogenic (extremities, breast) risk radiation therapy: Routine prophylaxis **not** recommended; however, for radiation therapy to the head and neck, thorax, or pelvis, or for minimal emetogenic risk radiation therapy, may use as rescue therapy (Ref).

IV (off label): 1 mg or 10 **mcg**/kg (Ref).

Oral: 2 mg (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

IV, Oral, Transdermal: No dosage adjustment necessary.

SUBQ (ER injection):

CrCl ≥60 mL/minute: No dosage adjustment necessary.

CrCl 30 to 59 mL/minute: 10 mg on day 1 of chemotherapy; do not administer more frequently than once every 14 days.

CrCl <30 mL/minute: Avoid use.

Dosing: Liver Impairment: Adult

Total clearance may be reduced by ~50% in patients with hepatic impairment. However, inter-subject variability limits interpretation of kinetic studies.

IV, Oral, Transdermal: No dose adjustment necessary (standard doses are well tolerated).

SubQ (ER injection): There is no dosage adjustment provided in the manufacturer's labeling.

Dosing: Adjustment for Toxicity: Adult

Serotonin syndrome: If serotonin syndrome occurs, discontinue 5-HT_3 receptor antagonist treatment and begin supportive management.

Transdermal patch: Skin reaction, severe or generalized (allergic rash, including erythematous, macular, or papular rash or pruritus): Remove patch.

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Granisetron: Pediatric drug information")

Chemotherapy-induced nausea and vomiting, prevention

Chemotherapy-induced nausea and vomiting (CINV), prevention:

Pediatric Oncology Group of Ontario (POGO) guidelines (Ref): **Note:** POGO guidelines do not recommend a maximum dose as most studies did not cap the dose (Ref).

Highly emetogenic chemotherapy:

Infants <6 months: IV: 40 mcg/kg as a single daily dose prior to chemotherapy; used in combination with dexamethasone.

Infants ≥6 months, Children, and Adolescents: IV: 40 mcg/kg as a single daily dose prior to chemotherapy; used in combination with dexamethasone and oral aprepitant/IV fosaprepitant (if no known or suspected drug interactions).

Moderately emetogenic chemotherapy:

Infants, Children, and Adolescents:

IV: 40 mcg/kg as a single daily dose prior to chemotherapy; used in combination with dexamethasone. For patients ≥6 months of age who cannot receive dexamethasone, use granisetron in combination with oral aprepitant/IV fosaprepitant (if no known or suspected drug interactions).

Oral: 40 mcg/kg/dose every 12 hours; used in combination with dexamethasone on chemotherapy days. For patients ≥6 months of age who cannot receive dexamethasone, use granisetron in combination with oral aprepitant/IV fosaprepitant (if no known or suspected drug interactions).

Low-emetogenic chemotherapy:

Infants, Children, and Adolescents:

IV: 40 mcg/kg as a single daily dose prior to chemotherapy.

Oral: 40 mcg/kg/dose every 12 hours on chemotherapy days.

Postoperative nausea and vomiting, prevention

Postoperative nausea and vomiting, prevention: Limited data available: Children and Adolescents: IV: 40 mcg/kg as a single dose; maximum dose: 0.6 mg/dose (Ref); **Note:** QT prolongation has been observed at this dose in pediatric patients 2 to 16 years of age; monitor closely.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

Children ≥2 years and Adolescents: IV, Oral: No dosage adjustment necessary.

Dosing: Liver Impairment: Pediatric

Children ≥2 years and Adolescents: IV, Oral: Pharmacokinetic studies in patients with hepatic impairment showed that total clearance was approximately halved; however, standard doses were very well tolerated, and dose adjustments are not necessary.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Reported incidences for adverse reactions are for the oral dosage form unless otherwise indicated.

>10%:

Gastrointestinal: Constipation (IV, oral, transdermal: 3% to 22%; including fecal impaction), nausea (20%), vomiting (12%)

Local: Injection-site reaction (IV: 37% to 62%; including bleeding at injection site [2% to 4%], bruising at injection site [≤45%], erythema at injection site [11% to 17%], hematoma at injection site [≤45%], induration at injection site [≤10%], injection-site infection [≤1%], injection-site nodule [≤18%], injection-site pruritus [≤2%], injection-site scarring [≤2%], irritation at injection site [≤2%], lipohypertrophy at injection site [≤2%], localized vesiculation [≤2%], pain at injection site [3% to 20%], paresthesia [injection site: ≤2%], rash at injection site [≤2%], residual mass at injection site [≤18%], skin discoloration at injection site [≤2%], swelling at injection site [≤10%], tenderness at injection site [4% to 27%], warm sensation at injection site [≤2%])

Nervous system: Asthenia (IV: 2% to 5%; oral: 14% to 18%), fatigue (IV: 11% to 21%), headache (IV, oral: 9% to 21%; transdermal: 1%)

1% to 10%:

Cardiovascular: Atrial fibrillation (IV: <3%), flushing (IV: <3%), hypertension (IV, oral: 1% to 2%), prolonged QT interval on ECG (IV, oral, transdermal: 1% to 3%; >450 milliseconds), syncope (IV: <3%)

Dermatologic: Alopecia (3%), skin rash (IV: 1%)

Gastrointestinal: Abdominal pain (IV, oral: 4% to 7%), decreased appetite (6%), diarrhea (IV, oral: 4% to 9%), dysgeusia (IV: 2%), dyspepsia (IV, oral: 3% to 6%), gastroesophageal reflux disease (IV: 1% to 5%), pancreatitis (IV: <3%)

Hematologic & oncologic: Anemia (4%), leukopenia (9%), thrombocytopenia (2%)

Hepatic: Increased serum alanine aminotransferase (IV, oral: >2 × ULN: 3% to 6%), increased serum aspartate aminotransferase (IV, oral: >2 × ULN: 3% to 5%)

Hypersensitivity: Hypersensitivity reaction (IV: <3%; including anaphylaxis)

Nervous system: Agitation (IV: <2%), anxiety (IV, oral: ≤2%), central nervous system stimulation (IV: <2%), dizziness (IV, oral: 3% to 5%), drowsiness (IV, oral: ≤4%), insomnia (IV, oral: 4% to 5%)

Miscellaneous: Fever (IV, oral: 3% to 9%)

Frequency not defined (any route of administration): Cardiovascular: Angina pectoris, atrioventricular block, cardiac arrhythmia, ECG abnormality, hypotension, sinus bradycardia, ventricular ectopy (including nonsustained ventricular tachycardia)

Postmarketing (any route of administration):

Cardiovascular: Asystole (Ref), bradycardia (Ref), chest pain, palpitations, sick sinus syndrome

Local: Application-site reaction (transdermal: application-site burning, application-site erythema, application-site irritation, application-site pain, application-site pruritus, application-site rash, application-site vesicles, skin discoloration [application site], urticaria [application site])

Contraindications

Hypersensitivity to granisetron or any component of the formulation or to other 5-HT3 receptor antagonists.

Canadian labeling: Additional contraindications (not in the US labeling): Concomitant use with apomorphine.

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Warnings/Precautions

Concerns related to adverse effects:

• ECG effects: Selective 5-HT_3 antagonists, including granisetron, have been associated with dose-dependent increases in ECG intervals (eg, PR, QRS duration, QT/QTc, JT), usually occurring with IV formulations (compared to oral) and occurring 1 to 2 hours after IV administration. In general, these changes are not clinically relevant; however, when used in conjunction with other agents that prolong these intervals (eg, Class I and III antiarrhythmics), arrhythmia or clinically relevant QT interval prolongation may occur resulting in torsade de pointes. Use with caution in patients at risk of QT prolongation and/or ventricular arrhythmia; patients with cardiac disease, electrolyte abnormalities, or who are receiving concomitant cardiotoxic chemotherapy are at higher risk.

• GI effects: Constipation may occur with all formulations, although a higher incidence is observed with tablets and the ER SUBQ injection. Hospitalization due to constipation or fecal impaction has been reported with the ER SUBQ injection. Progressive ileus and/or gastric distention may be masked by the ER SUBQ injection and transdermal patch (assess risks/benefits in patients with recent abdominal surgery). Granisetron does not stimulate gastric or intestinal peristalsis; do not use it in place of nasogastric suction.

• Hypersensitivity: Hypersensitivity reactions (including anaphylaxis) have been reported with granisetron in patients who have experienced hypersensitivity to other 5-HT_3 antagonists (cross-reactivity has been reported). Due to the ER properties of the SUBQ formulation, granisetron exposure may continue for 5 to 7 days following administration; hypersensitivity reactions may occur up to 7 days or longer following administration and may have an extended course.

• Injection site reactions: Serious or severe injection site reactions, including prolonged bleeding, bruising, hematomas, infections (eg, abscess, cellulitis with gangrene), nodules, pain, and tenderness, have occurred with the SUBQ ER formulation; may occur up to ≥2 weeks after administration. Some patients required hospitalization, IV antibiotics, incision/drainage, and/or surgical debridement. Patients with neutropenia or receiving concomitant anticoagulants or antiplatelet medication may be at increased risk. If injection site reactions occur, administer at a site away from the affected area; consider discontinuing in patients with severe or persistent reactions.

• Serotonin syndrome: Serotonin syndrome (SS) has been reported with 5-HT_3 receptor antagonists, predominantly when used in combination with other serotonergic agents (eg, selective serotonin reuptake inhibitors, serotonin and norepinephrine reuptake inhibitors, monoamine oxidase inhibitors, mirtazapine, fentanyl, lithium, tramadol, methylene blue). Some of the cases have been fatal. The majority of SS reports due to 5-HT_3 receptor antagonist have occurred in a postanesthesia setting or in an infusion center. SS has also been reported following overdose of another 5-HT_3 receptor antagonist. Signs of SS include mental status changes (eg, agitation, hallucinations, delirium, coma); autonomic instability (eg, tachycardia, labile BP, diaphoresis, dizziness, flushing, hyperthermia); neuromuscular changes (eg, tremor, rigidity, myoclonus, hyperreflexia, incoordination); GI symptoms (eg, nausea, vomiting, diarrhea); and/or seizures.

Disease-related concerns:

• Long QT syndrome: Use with caution in patients with congenital long QT syndrome or other risk factors for QT prolongation (eg, medications known to prolong QT interval, electrolyte abnormalities [hypokalemia or hypomagnesemia], cumulative high-dose anthracycline therapy).

Dosage form specific issues:

• Benzyl alcohol and derivatives: Some dosage forms may contain benzyl alcohol; large amounts of benzyl alcohol (≥99 mg/kg/day) have been associated with a potentially fatal toxicity ("gasping syndrome") in neonates; the "gasping syndrome" consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension, and cardiovascular collapse (AAP ["Inactive" 1997]; CDC 1982); some data suggest that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol with caution in neonates. See manufacturer's labeling.

• Polysorbate 80: Some dosage forms may contain polysorbate 80 (also known as Tweens). Hypersensitivity reactions, usually a delayed reaction, have been reported following exposure to pharmaceutical products containing polysorbate 80 in certain individuals (Isaksson 2002; Lucente 2000; Shelley 1995). Thrombocytopenia, ascites, pulmonary deterioration, and renal and hepatic failure have been reported in premature neonates after receiving parenteral products containing polysorbate 80 (Alade 1986; CDC 1984). See manufacturer's labeling.

• Transdermal patch: Application-site reactions have occurred with use; local reactions were generally mild and did not require discontinuation. Granisetron patch may potentially be affected by natural or artificial sunlight.

Other warnings/precautions:

• Chemotherapy-associated emesis: Antiemetics are most effective when used prophylactically (MASCC/ESMO [Herrstedt 2024]). If emesis occurs despite optimal antiemetic prophylaxis, reevaluate emetic risk, disease status, concurrent morbidities and current medications to assure antiemetic regimen is optimized (ASCO [Hesketh 2020]).

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Patch, Transdermal:

Sancuso: 3.1 mg/24 hr (1 ea)

Prefilled Syringe, Subcutaneous:

Sustol: 10 mg/0.4 mL (0.4 mL) [contains polyethylene glycol (macrogol)]

Solution, Intravenous:

Generic: 1 mg/mL (1 mL); 4 mg/4 mL (4 mL)

Solution, Intravenous [preservative free]:

Generic: 1 mg/mL (1 mL)

Solution, Oral:

Granisol: 2 mg/10 mL (30 mL) [contains fd&c yellow #6 (sunset yellow), sodium benzoate; orange flavor]

Tablet, Oral:

Generic: 1 mg

Generic Equivalent Available: US

May be product dependent

Pricing: US

Patch (Sancuso Transdermal)

3.1 mg/24 hrs (per each): $895.74

Prefilled Syringe (Sustol Subcutaneous)

10 mg/0.4 mL (per 0.4 mL): $997.08

Solution (Granisetron HCl Intravenous)

1 mg/mL (per mL): $10.80 - $24.00

4 mg/4 mL (per mL): $7.50 - $10.80

Solution (Granisol Oral)

2 mg/10 mL (per mL): $74.00

Tablets (Granisetron HCl Oral)

1 mg (per each): $59.01 - $59.05

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous:

Generic: 1 mg/mL (1 mL, 4 mL)

Tablet, Oral:

Generic: 1 mg

Extemporaneous Preparations

0.2 mg/mL Oral Suspension

A 0.2 mg/mL oral suspension may be made with tablets. Crush twelve 1 mg tablets in a mortar and reduce to a fine powder. Add 30 mL distilled water, mix well, and transfer to a bottle. Rinse the mortar with 10 mL cherry syrup and add to bottle. Add sufficient quantity of cherry syrup to make a final volume of 60 mL. Label "shake well". Stable 14 days at room temperature or refrigerated (Ref).

0.05 mg/mL Oral Suspension

A 0.05 mg/mL oral suspension may be made with tablets and one of three different vehicles (Ora-Sweet, Ora-Plus, or a mixture of methylcellulose 1% and Simple Syrup, N.F.). Crush one 1 mg tablet in a mortar and reduce to a fine powder. Add 20 mL of the chosen vehicle and mix to a uniform paste; transfer to a calibrated bottle. Label ”shake well” and ”refrigerate”. Stable for 91 days refrigerated (Ref).

Administration: Adult

Oral: Doses should be administered up to 1 hour prior to initiation of chemotherapy/radiation.

IV: Administer IV push over 30 seconds or as a 5-minute infusion.

SUBQ (ER injection): For SUBQ administration only. Administer as a single injection at the back of the upper arm or in abdomen (at least 1 inch away from the umbilicus). Avoid using areas where the skin is burned, hardened, inflamed, swollen, or otherwise compromised. Inject slowly; may take up to 20 to 30 seconds (product is viscous, pressing the syringe plunger will not result in faster expulsion). A topical anesthetic may be applied at injection site prior to administration. Do not administer if particulate matter or discoloration is observed, if the tip cap is missing or has been tampered with, or the luer fitting is missing or dislodged. Remove from refrigerator at least 60 minutes prior to administration; remove from pack and activate 1 syringe warming pouch and wrap syringe with warming pouch for 5 to 6 minutes to allow warming to body temperature. Injection site reactions may occur; if injection site reaction is not yet resolved prior to the next dose, rotate injection site with next administration.

Transdermal (Sancuso): Apply patch by pressing down firmly to clean, dry, nearly hairless, intact skin on upper outer arm; smooth down with fingers. Wash hands thoroughly after applying. Do not shave hair where applying; clip hair as close to the skin as possible. Do not use on red, irritated, or damaged skin. Do not apply to skin where creams, oils, lotions, powders, or other skin products have been applied; may prevent patch from adhering properly. Remove patch from pouch immediately before application. Do not cut patch. Cover patch application site with clothing to protect from natural or artificial sunlight exposure while patch is applied and for 10 days following removal; granisetron may potentially be affected by natural or artificial sunlight. Do not apply heat (eg, heating pad, heating lamp) over or in area of the transdermal patch; avoid prolonged exposure to heat (may increase plasma concentrations). Do not apply more than 1 patch at a time. Surgical or medical adhesive tape may be applied to lifted edges of patch to keep in place; do not completely cover patch or wrap tape around arm. Contact with water while bathing or showering will not affect patch; however, avoid swimming, saunas, hot tubs, and strenuous exercise. Dispose of any used or unused patches by folding adhesive ends together and discard properly in trash away from children and pets.

Enteral feeding tube:

The following recommendations are based upon the best available evidence and clinical expertise. Senior editorial team: Joseph I. Boullata, PharmD, RPh, CNS-S, FASPEN, FACN; Peggi A. Guenter, PhD, RN, FASPEN; Kathleen Gura, PharmD, BCNSP, FASHP, FASPEN, FPPA, FMSHP; Mark G. Klang, MS, RPh, BCNSP, PhD, FASPEN; Linda Lord, NP, ACNP-BC, CNSC, FASPEN; Lucas E. Orth, PharmD, BCPPS; Russel J. Roberts, PharmD, BCCCP, FCCM.

Oral tablet:

Gastric (eg, NG, G-tube) or post-pyloric (eg, J-tube) tubes (≥8 French): Crush tablet(s) into a fine powder and disperse in 10 mL purified water immediately prior to administration; draw up mixture into enteral dosing syringe and administer via feeding tube (Ref). Alternatively, may place intact tablet into the barrel of an empty enteral dosing syringe, draw 10 mL of purified water into the syringe, and allow tablet to disperse; if necessary, shake syringe (~5 minutes) to disperse and then immediately administer via feeding tube (Ref).

Dosage form information: Some formulations may be film-coated; administration of film-coated granisetron tablets via feeding tube may increase the risk of clogging the tube; if used, ensure tablets are dispersed sufficiently with an adequate amount of purified water prior to administration (Ref).

General guidance: Hold enteral nutrition (EN) during granisetron administration (Ref). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) before administration (Ref). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose (Ref). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) and restart EN (Ref).

Note : Recommendations may not account for differences in inactive ingredients, osmolality, or other formulation properties that may vary among products from different manufacturers.

Preparation for Administration: Adult

IV: May be administered undiluted or may be further diluted in NS or D5W.

SUBQ (ER injection): Remove ER injection kit from refrigeration at least 60 minutes prior to administration. Allow the syringe and all other contents to warm to room temperature. Activate 1 syringe warming pouch and wrap the syringe in the warming pouch for 5 to 6 minutes to allow to reach body temperature. Do not substitute non-kit components for any components of the administration kit.

Administration: Pediatric

The following feeding tube recommendations are based upon the best available evidence and clinical expertise. Senior editorial team: Joseph I. Boullata, PharmD, RPh, CNS-S, FASPEN, FACN; Peggi A. Guenter, PhD, RN, FASPEN; Kathleen Gura, PharmD, BCNSP, FASHP, FASPEN, FPPA, FMSHP; Mark G. Klang, MS, RPh, BCNSP, PhD, FASPEN; Linda Lord, NP, ACNP-BC, CNSC, FASPEN; Lucas E. Orth, PharmD, BCPPS; Russel J. Roberts, PharmD, BCCCP, FCCM.

Note: Recommendations may not account for differences in inactive ingredients, osmolality, or other formulation properties that may vary among products from different manufacturers.

Oral: Administer 30 minutes to 1 hour prior to chemotherapy and repeat in 12 hours (Ref).

Tablet:

Administration via feeding tube:

Gastric (eg, NG, G-tube) or post-pyloric (eg, J-tube) tubes (≥8 French): Crush tablet(s) into a fine powder and disperse in 10 mL purified water immediately prior to administration; draw up mixture into enteral dosing syringe and administer via feeding tube (Ref). Alternatively, may place intact tablet into the barrel of an empty enteral dosing syringe, draw 10 mL of purified water into the syringe, and allow tablet to disperse; if necessary, shake syringe (~5 minutes) to disperse and then immediately administer via feeding tube (Ref).

Dosage form information: Some formulations may be film-coated; administration of film-coated granisetron tablets via feeding tube may increase the risk of clogging the tube; if used, ensure tablets are sufficiently dispersed prior to administration (Ref).

General guidance: Hold enteral nutrition during granisetron administration (Ref). Flush feeding tube with the lowest volume of purified water necessary to clear the tube prior to administration based on size of patient and/or feeding tube (eg, neonates: 1 to 3 mL; infants and children: 2 to 5 mL; adolescents: 15 mL); refer to institutional policies and procedures (Ref). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose (Ref). Flush feeding tube with an appropriate volume of purified water and restart enteral nutrition (Ref).

Parenteral: IV: Infuse over 30 seconds undiluted **or** diluted and administer over 5 minutes; infusions administered over 30 to 60 minutes have also been reported (Ref).

Preparation for Administration: Pediatric

Parenteral: IV: May be diluted in a small volume of NS or D5W; volumes ranging from 2 to 20 mL have been reported in trials with pediatric patients (Ref)

Storage/Stability

IV: Store at 20°C to 25°C (68°F to 77°F); excursions permitted between 15°C to 30°C (59°F to 86°F). Protect from light. Do not freeze. After initial entry, use contents of multidose vial within 30 days.

Stable when mixed in NS or D5W for at least 24 hours at room temperature. **Note:** Storage recommendations may differ for products available in Canada; refer to product labeling.

Oral: Store at 15°C to 30°C (59°F to 86°F). Protect from light.

SUBQ (ER injection): Store at 2°C to 8°C (36°F to 46°F); do not freeze. May be placed back in refrigerator after being kept at room temperature; may remain at room temperature for a maximum of 7 days. Protect from light.

Transdermal patch: Store at 20°C to 25°C (68°F to 77°F). Keep patch in original packaging until immediately prior to use.

Use: Labeled Indications

Chemotherapy-associated nausea and vomiting: Prevention of nausea and vomiting associated with initial and repeat courses of emetogenic chemotherapy, including high-dose cisplatin (injection and tablets); prevention of nausea and vomiting associated with anthracycline/cyclophosphamide chemotherapy regimens; prevention of nausea and vomiting associated with moderately and/or highly emetogenic chemotherapy regimens of up to 5 consecutive days of duration (transdermal).

Postoperative nausea and vomiting: Prevention and treatment of postoperative nausea and vomiting in adults.

Radiation-associated nausea and vomiting: Prevention of nausea and vomiting associated with radiation therapy, including total body radiation and fractionated abdominal radiation (tablets).

Medication Safety Issues

Sound-alike/look-alike issues:

Granisetron may be confused with dolasetron, ondansetron, palonosetron

Metabolism/Transport Effects

Substrate of CYP3A4 (Minor); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Apomorphine: Antiemetics (5HT3 Antagonists) may increase hypotensive effects of Apomorphine. *Risk X: Avoid*

Haloperidol: QT-prolonging Agents (Indeterminate Risk - Caution) may increase QTc-prolonging effects of Haloperidol. *Risk C: Monitor*

QT-prolonging Agents (Highest Risk): QT-prolonging Agents (Indeterminate Risk - Caution) may increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Serotonergic Agents (High Risk): Antiemetics (5HT3 Antagonists) may increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

TraMADol: Antiemetics (5HT3 Antagonists) may increase serotonergic effects of TraMADol. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Pregnancy Considerations

Adverse events have not been observed in animal reproduction studies. In an ex vivo placental perfusion study, granisetron was shown to cross the placenta in a concentration (dose) dependent manner (Julius 2014). Initial studies note the pharmacokinetics of the transdermal system may be different in pregnant women. A relationship between granisetron plasma concentrations and relief of symptoms of nausea and vomiting of pregnancy was also observed (Caritis 2016). Some dosage forms (injection) may contain benzyl alcohol.

Breastfeeding Considerations

It is not known if granisetron is excreted in breast milk.

According to the manufacturer, the decision to continue or discontinue breast-feeding during therapy should take into account the risk of infant exposure, the benefits of breast-feeding to the infant, benefits of treatment to the mother, and the underlying maternal condition.

Monitoring Parameters

Monitor for development of constipation and for decreased bowel activity (particularly in patients at risk for GI obstruction). Monitor for signs/symptoms of hypersensitivity. Monitor for signs of serotonin syndrome.

ER SUBQ injection: Monitor for injection site reactions (eg, prolonged bleeding, bruising, hematomas, infections [abscess, cellulitis with gangrene], nodules, pain, and tenderness) for at least 2 weeks after administration.

Mechanism of Action

Selective 5-HT_3-receptor antagonist, blocking serotonin, both peripherally on vagal nerve terminals and centrally in the chemoreceptor trigger zone

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: IV: 1 to 3 minutes.

Duration: Oral, IV: Generally up to 24 hours; SUBQ (extended release): Remains detectable in the plasma for 7 days.

Absorption: Transdermal patch: ~66% over 7 days.

Distribution: V_d: 2 to 4 L/kg; widely throughout body.

Protein binding: ~65%.

Metabolism: Hepatic via CYP1A1 and CYP3A4 N-demethylation, oxidation, and conjugation; some metabolites may have 5-HT_3 antagonist activity.

Half-life elimination: Oral: 6 hours; IV: Mean range: 5 to 9 hours; SUBQ (extended release): ~24 hours.

Time to peak, plasma: Transdermal patch: Maximum systemic concentrations: ~48 hours after application (range: 24 to 168 hours); SUBQ (extended release): ~24 hours.

Excretion: Urine (11% to 12% as unchanged drug, 48% to 49% as metabolites); feces (34% to 38% as metabolites).

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Hepatic function impairment: Total clearance is decreased by about 50% in patients with hepatic impairment (due to neoplastic disease) compared to patients with normal hepatic function.

Older adult: Mean values were lower for Cl and longer for half-life.

Sex: Men had a higher C_{max}, but there was no difference in AUC.

Medication Guide and/or Vaccine Information Statement (VIS)

An FDA-approved patient medication guide, which is available with the product information and as follows, must be dispensed with this medication:

Sustol: https://www.accessdata.fda.gov/drugsatfda_docs/label/2026/022445s014lbl.pdf#page=17

Patient Counseling Points

(For additional information see ”Granisetron: Patient drug information”)

What is this drug used for?

• It is used to prevent upset stomach and throwing up.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

All products:

• Headache

• Feeling dizzy, sleepy, tired, or weak

• Diarrhea or constipation

• Heartburn

• Decreased appetite

• Trouble sleeping

Skin patch:

• Irritation where this drug was used

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

All products:

• High or low blood pressure like very bad headache or dizziness, passing out, or change in eyesight

• Chest pain or pressure

• Fast or abnormal heartbeat

• Shortness of breath

• Stomach pain

• Swelling of belly

• Fever, chills, or sore throat; any unexplained bruising or bleeding; or feeling very tired or weak

• Serotonin syndrome like agitation; change in balance; confusion; hallucinations; fever; fast or abnormal heartbeat; flushing; muscle twitching or stiffness; seizures; shivering or shaking; sweating a lot; severe diarrhea, upset stomach, or throwing up; or very bad headache

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat

Long-acting injection:

• Injection site pain or tenderness that you need to take a pain drug for or that causes problems with daily living

• Area that feels hard or bruise at the injection site that does not go away

• Infection at the injection site like redness, swelling, warmth, oozing, or skin color change

• Bleeding at the injection site that is very bad or lasts longer than 24 hours

• Constipation or if constipation gets worse after you use this drug

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Grani denk | Kytril | Sancuso;
- (AR) Argentina: Eumetic | Kytril | Rigmoz;
- (AT) Austria: Granisetron | Granisetron actavis | Granisetron b. braun | Granisetron kabi | Granitron | Kytril | Sancuso;
- (AU) Australia: Apo granisetron | Granisetron aft | Granisetron apotex | Granisetron kabi | Kytril;
- (BD) Bangladesh: Naurif;
- (BE) Belgium: Granisetron fresenius | Granisetron Ips | Kytril | Sancuso;
- (BF) Burkina Faso: Grani denk;
- (BG) Bulgaria: Granegis | Granisetron Teva | Kytril | Rasetron;
- (BR) Brazil: Cloridrato de granisetrona | Grandax | Kytril | Neosetron;
- (CH) Switzerland: Granisetron fresenius | Granisetron labatec iv | Kytril;
- (CI) Côte d'Ivoire: Gramy ae;
- (CL) Chile: Granisetron | Kytril;
- (CN) China: A pa | A si mi ya | Ai bang | Ai ting | Ba tai | Bai hong | Bi li | Di kang li shu | Dong bao ou ke ning | Ge la ke | Ge nai ya | Ge ou shu | Ge rui tong | Ge xin | Granisetron | Granisetron hydrochloride and glucose | Granisetron hydrochloride and sodium chloride | Jie dan | Kytril | Li sheng an | Ou zhi ning | Qiong sha ao | Run dan | Shu er zhi | Shu xing | Si li mei nuo | Si nuo kang xin | Tian quan kang xin | Tu ting | Wei kun | Zhongbao yige;
- (CO) Colombia: Cetrexon | Kytril;
- (CR) Costa Rica: Grani denk | Granisetron | Granisetron clorhidrato | Kytril;
- (CZ) Czech Republic: Emegar | Granegis | Granisetron b. braun | Granisetron kabi | Granisetron Sandoz | Granisetron Teva | Grateva | Kytril;
- (DE) Germany: Axigran | Grani denk | Granisetron | Granisetron actavis | Granisetron B.Braun | Granisetron denk | Granisetron Dura | Granisetron Haemato | Granisetron Hameln | Granisetron Hexal | Granisetron hikma | Granisetron kabi | Granisetron puren | Granisetron ratiopharm | Granisetron rotexmedica | Granisetron stada | Granisetron Teva | Kevatril | Kytril | Ribosetron | Sancuso;

- (DO) Dominican Republic: Granicip | Kytril;
- (EC) Ecuador: Granisetron | Granisetron clorhidrato | Kytril;
- (EE) Estonia: Granisetron Bmm | Granisetron Teva | Granissetrom aurovitas | Kytril;
- (EG) Egypt: Em ex | Emetovoid | Eupivora | G setron | Granitryl | Kytril | Mitashe;
- (ES) Spain: Granisetron actavis | Granisetron Combino Pharm | Granisetron G.E.S. | Granisetron kabi | Granisetron Normon | Granisetron Sandoz | Granisetron Teva | Kytril;
- (ET) Ethiopia: Grani denk;
- (FI) Finland: Granisetron Bmm | Granisetron Braun | Granisetron Claris | Granisetron fresenius kabi | Granisetron Hameln | Granisetron Sandoz | Granisetron stada | Granisetron Teva | Granisetron Vianex | Kytril | Sancuso;
- (FR) France: Granisetron b braun | Granisetron Dci | Granisetron Teva | Kytril;
- (GB) United Kingdom: Granisetron | Kytril | Sancuso;
- (GR) Greece: Granisetron | Granisetron actavis | Kytril | Sancuso;
- (GT) Guatemala: Grani denk | Kytril;
- (HK) Hong Kong: Granisetron | Granisetron kabi | Kytril | Sancuso;
- (HR) Croatia: Granisetron B.Braun | Granisetron kabi | Kytril;
- (HU) Hungary: Granegis | Granigen | Granisetron actavis | Granisetron kabi | Granisetron Pharmacenter | Kytril;
- (ID) Indonesia: Granisetron | Granon | Granopi | Kytril;
- (IE) Ireland: Granisetron | Kytril;
- (IL) Israel: Kytril | Setron;
- (IN) India: Cadigran | Emegran | Grandem | Granicip | Granifer | Graniforce | Granirex | Graniset | Granitero | Graniz | Grantis;
- (IT) Italy: Grandise | Granisetron | Granisetron b. braun | Granisetron fki | Granisetron Mylan | Granisetron Teva Italia | Kytril | Pandiol | Sancuso;
- (JO) Jordan: Kytril | Setonix;
- (JP) Japan: Granisetron | Granisetron meiji | Granisetron sawai | Granisetron towa | Kytril;
- (KE) Kenya: Grani denk | Graniset | Granisetron martindale pharma | Kytril | Setonix;
- (KR) Korea, Republic of: Gratril | Kabi granisetron | Kanitron | Kyseron | Kytril | Sancuso;
- (KW) Kuwait: Sancuso;
- (LB) Lebanon: Grani denk | Kytril | Sancuso;
- (LT) Lithuania: Granicip | Granisetron | Granisetron Bmm | Granisetron Teva | Granitero | Kytril;
- (LU) Luxembourg: Granisetron Ips | Kytril;
- (LV) Latvia: Granisetron Teva | Kytril;
- (MA) Morocco: Kytril;
- (MX) Mexico: Gavazetrin | Granisetron | Granisetron tecnofarma | Kogra | Kytril | Manotrox | Tentratec | Trilemes | Vogracen;
- (MY) Malaysia: Granisetron | Granodex | Kytril | Kytron | Sentroni | Vaxcel granisetron | Viatrinil;
- (NL) Netherlands: Granisetron | Granisetron actavis | Granisetron cf | Granisetron fresenius kabi | Granisetron Hameln | Granisetron merck | Granisetron Sandoz | Kytril | Sancuso;
- (NO) Norway: Granisetron | Granisetron algol | Granisetron Hameln | Kytril | Sancuso;
- (NZ) New Zealand: Granirex | Granisetron;
- (PA) Panama: Grani denk | Granisetron;
- (PE) Peru: Ganiset | Granisetron | Helminar | Kytril | Suliftop;
- (PH) Philippines: Grakytil | Kytril | Navom | Sancuso;
- (PK) Pakistan: Aniset | Graniset | Gratron | Kytril | Sitensa;
- (PL) Poland: Kytril;
- (PR) Puerto Rico: Granisetron HCL | Kytril | Sancuso | Sustol;
- (PT) Portugal: Granisetron actavis | Granisetron combino | Granisetron generis | Granisetron hikma | Granisetron kabi | Granisetron Teva | Granissetrom hikma | Kytril;
- (QA) Qatar: Kytril | Sancuso;
- (RO) Romania: Granegis | Granisetron actavis | Granisetron kabi | Granisetron Teva | Granored | Kytril | Sancuso;
- (RU) Russian Federation: Avomit | Granisetron | Granisetron kabi | Granisetron Teva | Granisetron TL | Granisetrone | Kitril | Kytril | Notirol;
- (SA) Saudi Arabia: Granisetron | Granisetron kabi | Granitron | Kytril | Sancuso;

- (SE) Sweden: Granisetron b braun | Granisetron bmm pharma | Granisetron fresenius kabi | Granisetron Hameln | Granisetron stada | Kytril | Sancuso;
- (SG) Singapore: Granisetron aft | Kytril | Sancuso;
- (SI) Slovenia: Granisetron | Granisetron lek | Granisetron Teva | Kytril;
- (SK) Slovakia: Emegar | Granegis | Granisetron kabi | Granisetron Teva | Kytril | Rasetron;
- (TH) Thailand: Kytril;
- (TN) Tunisia: Sancuso;
- (TR) Turkey: Emetril | Granexa | Granidur | Granitron | Granset | Gratryl | GRONSIT | Kytril | Neoset | Setrex | Setron | Sinarex | Tigron | Vitasetron;
- (TW) Taiwan: Granisetron kabi | Grantron | Kytril | Otril | Setron | Unvomit | Vomstop;
- (UY) Uruguay: Eumetic | Granisetron | Kytril;
- (VE) Venezuela, Bolivarian Republic of: Kytril | Rubrum;
- (VN) Viet Nam: Bfs grani | Rapogy;
- (ZA) South Africa: ADCO Granisetron | Aspen Granisetron | Granicip | Granisetron fresenius | Granisetron Teva | Granitril | Grantryl | Kytril | Mylan Granisetron | Sandoz Granisetron;
- (ZM) Zambia: Grani denk

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